

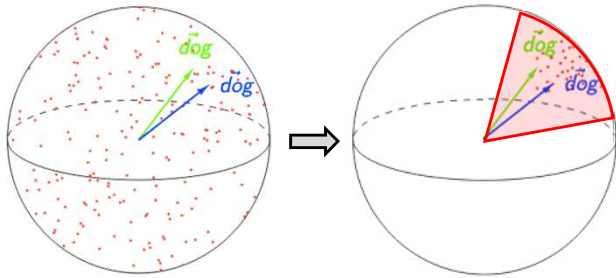
Debiased Contrastive learning of Unsupervised Sentence Representations

ACL, 2022

김재민

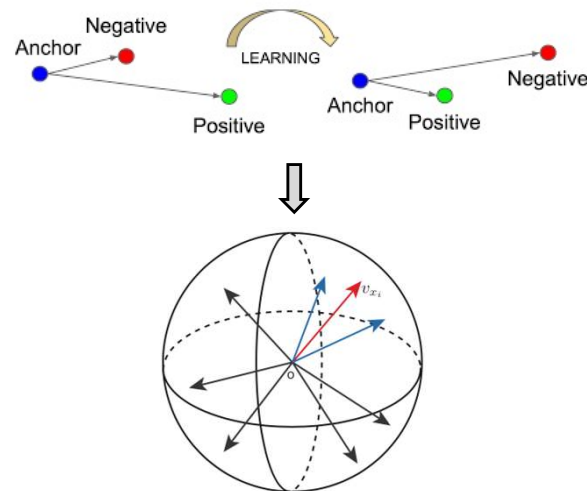
01 Background

Sentence representation learning



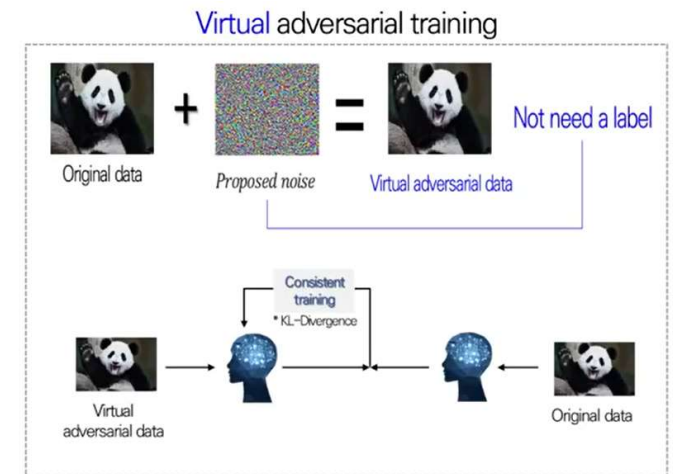
- ☑ Universal sentence representation
→ The key to the success of various task
- ☑ Recently, strong potential을 갖고 있는 PLM을 활용
→ 하지만, Anisotropy한 문제를 갖고 있음

Contrastive learning



- ☑ Alignment & Uniformity in semantic representation space
- ☑ Positive pair를 구성하기 위한 연구가 활발
→ 하지만, randomly sample negatives는 거의 없음

Virtual Adversarial Training

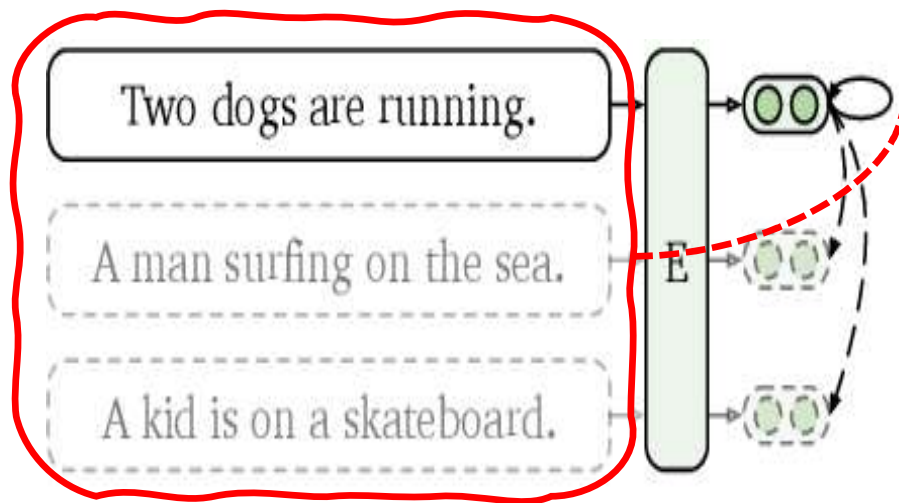


- ☑ NLP에서 adversarial perturbation이 embedding layer 에서 많이 이뤄짐
→ Text classification, machine translation, NLU task에서 효과적임을 보여줌

02 Introduction

□ SimCSE (EMNLP,2021)

Unsupervised SimCSE



- ☑ 동일한 문장을 input으로 넣어 Positive pair
- ☑ Dropout (noise) 활용한 minimal augmentation
- ☑ dropout 을 바꾸어 나온 hidden Representation을 positive pair 로 구성

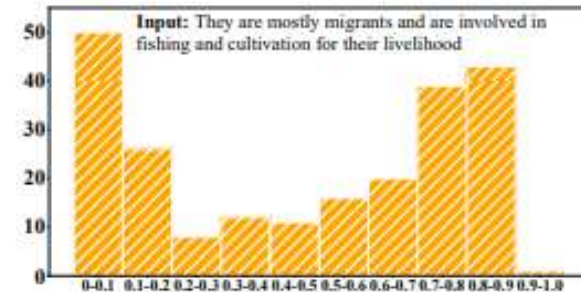


Figure 1: The distribution of cosine similarity between an input sentence and 255 in-batch negatives from the commonly-used Wikipedia Corpus. It is evaluated by the SimCSE model (Gao et al., 2021). Almost half of the negatives have high similarities with the input.

- ☑ in-batch negative sentence 들이 높은 similarit를 갖고 있음

논문에서는, 반 이상이 0.7이상의 cos similarity를 갖고 있음

- ☑ PLM의 Anisotropy 한 문제는 모든 semantics를 representation space에 uniformity하게 임베딩하기 어려움

02 Introduction

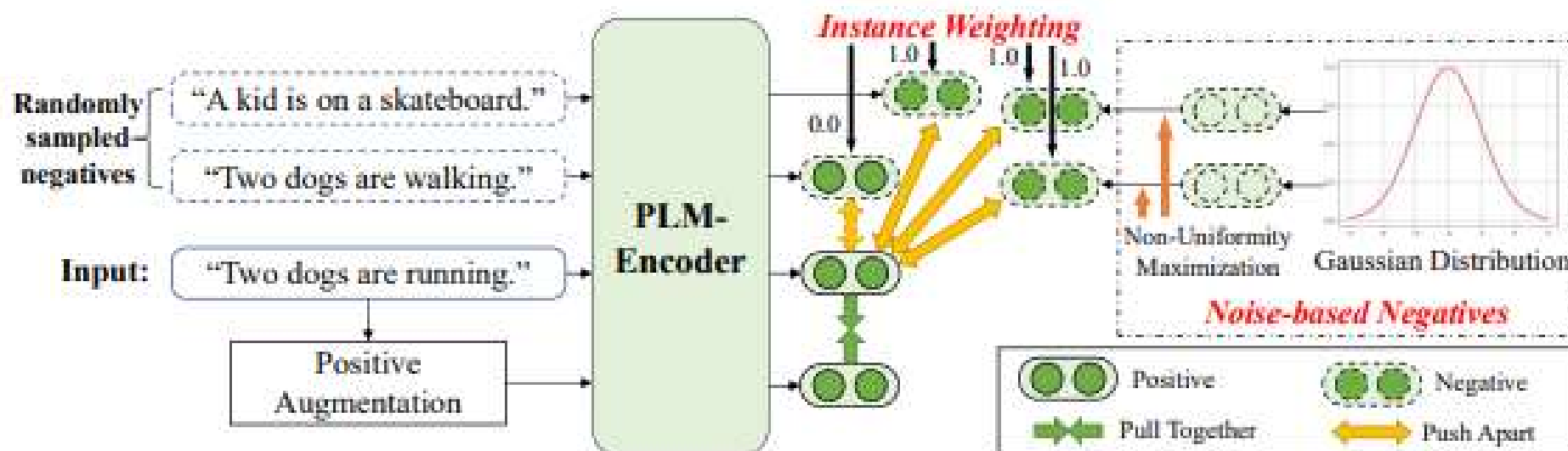


Figure 2: The overview of our DCLR framework with noise-based negatives and the instance weighting strategy. We show the case that a false negative is punished by assigning the weight 0.

□ DCLR (Debiased Contrastive Learning of unsupervised sentence Representation)

□ Core idea

1. Design an instance weighting method → punish the sampled **false negatives** during training
2. New negative based on random Gaussian noise, devise a gradient-based algorithm to optimize the noise-based negatives towards the most **nonuniform points** → improve the uniformity of the representation space

02 Introduction

(1) Our approach is the first attempt to reduce the sampling bias in contrastive learning of unsupervised sentence representation

(2) DCLR

(3) 실험에서 7개의 STS tasks 에서 효과적임을 보였다.

□ DCLR

(ation)

□ Core idea

1. Design an instance weighting method → punish the sampled false negatives during training
2. New negative based on random Gaussian noise, devise a gradient-based algorithm to optimize the noise-based negatives towards the most nonuniform points → improve the uniformity of the representation space

03 Approach

Generating Noise based Negatives

$$\{\hat{h}_1, \hat{h}_2, \dots, \hat{h}_k\} \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

noise-based negatives $\{\hat{h}_j\}$

(h_i, h_i^+) as:

$$L_U(h_i, h_i^+, \{\hat{h}\}) = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau_u}}{\sum_{\hat{h}_j \in \{\hat{h}\}} e^{\text{sim}(h_i, \hat{h}_j)/\tau_u}}, \quad (2)$$

$\text{sim}(h_i, h_i^+)$ is the cosine similarity $\frac{h_i^\top h_i^+}{\|h_i\| \cdot \|h_i^+\|}$.

$$\hat{h}_j = \hat{h}_j + \beta g(\hat{h}_j) / \|g(\hat{h}_j)\|_2, \quad (3)$$

$$g(\hat{h}_j) = \nabla_{\hat{h}_j} L_U(h_i, h_i^+, \{\hat{h}\}), \quad (4)$$

Contrastive Learning with Instance Weighting

$$\alpha_{h^-} = \begin{cases} 0, & \text{sim}_C(h_i, h^-) \geq \phi \\ 1, & \text{sim}_C(h_i, h^-) < \phi \end{cases} \quad (5)$$

$$L = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{h^- \in \{\hat{h}\} \cup \{\tilde{h}^-\}} \alpha_{h^-} \times e^{\text{sim}(h_i, h^-)/\tau}}, \quad (6)$$

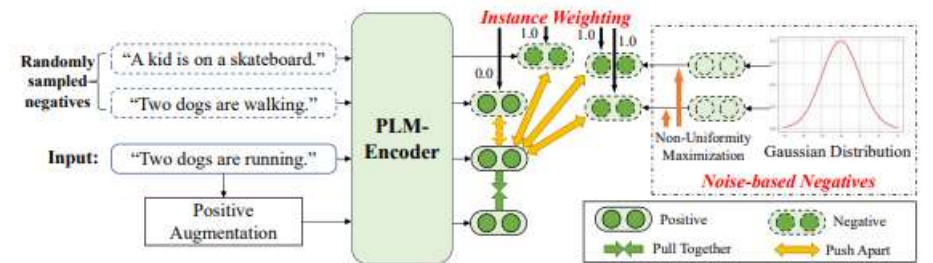


Figure 2: The overview of our DCLR framework with noise-based negatives and the instance weighting strategy. We show the case that a false negative is punished by assigning the weight 0.

04 Experiment

□ Baseline Methods (unsupervised sentence representation)

- Glove
- USE
- CLS, Mean, First-last AVG
- Flow
- Whitening
- Contrastive (BT)
- ConSERT
- SG-OPT
- SimCSE

Not
PLM

isotropy

CL

		STS12	STS13	STS14	STS15	STS16	STS-B	SICK-R	Avg.
Non-BERT	Glove(avg.)	55.14	70.66	59.73	68.25	63.66	58.02	53.76	61.32
	USE	64.49	67.8	64.61	76.83	73.18	74.92	76.69	71.22
BERT-base	CLS	21.54	32.11	21.28	37.89	44.24	20.3	42.42	31.4
	Mean	30.87	59.89	47.73	60.29	63.73	47.29	58.22	52.57
	First-Last AVG	39.7	59.38	49.67	66.03	66.19	53.87	62.06	56.7
	flow	58.4	67.1	60.85	75.16	71.22	68.66	64.47	66.55
	whitening	57.83	66.9	60.9	75.08	71.31	68.24	63.73	66.28
	Contrastive(BT)	54.26	64.03	54.28	68.19	67.5	63.27	66.91	62.63
	ConSERT	64.64	78.49	69.07	79.72	75.95	73.97	67.31	72.74
	SG-OPT	66.84	80.13	71.23	81.56	77.17	77.23	68.16	74.62
	SimCSE	68.4	82.41	74.38	80.91	78.56	76.85	72.23	76.25
	DCLR	70.81	83.73	75.11	82.56	78.44	78.31	71.59	77.22
BERT-large	CLS	27.44	30.76	22.59	29.98	42.74	26.75	43.44	31.96
	Mean	27.67	55.79	44.49	51.67	61.88	47	53.85	48.91
	First-Last AVG	57.73	61.17	61.18	68.07	70.25	59.59	60.34	62.62
	flow	62.82	71.24	65.39	78.98	73.23	72.72	63.77	70.07
	whitening	64.34	74.6	69.64	74.68	75.9	72.48	60.8	70.35
	Contrastive(BT)	52.04	62.59	54.25	71.07	66.71	63.84	66.53	62.43
	ConSERT	70.69	82.96	74.13	82.78	76.66	77.53	70.37	76.45
	SG-OPT	67.02	79.42	70.38	81.72	76.35	76.16	70.2	74.46
	SimCSE	70.88	84.16	76.43	84.5	79.76	79.26	73.88	78.41
	DCLR	71.87	84.83	77.37	84.7	79.81	79.55	74.19	78.9
RoBERTa-base	CLS	16.67	45.57	30.36	55.08	56.98	45.41	61.89	44.57
	Mean	32.11	56.33	45.22	61.34	61.98	54.53	62.03	53.36
	First-Last AVG	40.88	58.74	49.07	65.63	61.48	58.55	61.63	56.57
	whitening	46.99	63.24	57.23	71.36	68.99	61.36	62.91	61.73
	Contrastive(BT)	62.34	78.6	68.65	79.31	77.49	79.93	71.97	74.04
	SG-OPT	62.57	78.96	69.24	79.99	77.17	77.6	68.42	73.42
	SimCSE	70.16	81.77	73.24	81.36	80.65	80.22	68.56	76.57
	DCLR	70.01	83.08	75.09	83.66	81.06	81.86	70.33	77.87
RobBERTa-large	CLS	19.25	22.97	14.93	33.41	38.01	12.52	40.63	25.96
	Mean	33.63	57.22	45.67	63	61.18	47.07	58.38	52.31
	First-Last AVG	58.91	58.62	61.44	69.05	65.23	59.38	58.84	61.64
	whitening	64.17	73.92	71.06	76.4	74.87	71.68	58.49	70.08
	Contrastive(BT)	57.6	72.14	62.25	71.49	71.75	77.05	67.83	68.59
	SG-OPT	64.29	76.36	68.48	80.1	76.6	78.14	67.97	73.13
	SimCSE	72.86	83.99	75.62	84.77	81.8	81.98	71.26	78.9
	DCLR	73.09	84.57	76.13	85.15	81.99	82.35	71.8	79.3

04 Experiment

❑ Debiased Contrastive learning on Other Methods

Model	STS-Avg.
BERT-base+Ours	77.22
w/o Noise-based Negatives	76.17
w/o Instance Weighting	76.31
BERT-base+Random Noise	75.22
BERT-base+Knowledge Distillation	75.05
BERT-base+Self Instance Weighting	73.93

Table 2: Ablation and variation studies of our approach on the test set of seven STS tasks.

04 Experiment

❑ Debiased Contrastive learning on Other Methods

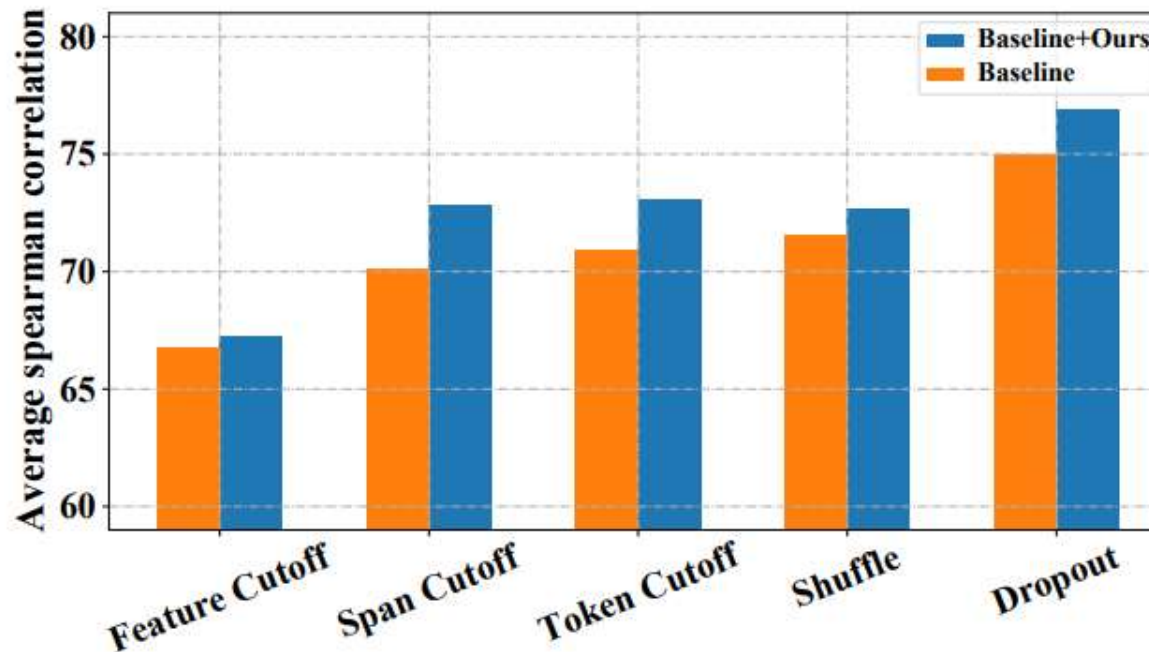


Figure 3: Performance comparison using different positive augmentation strategies on the test set of seven STS tasks.

04 Experiment

□ Uniformity

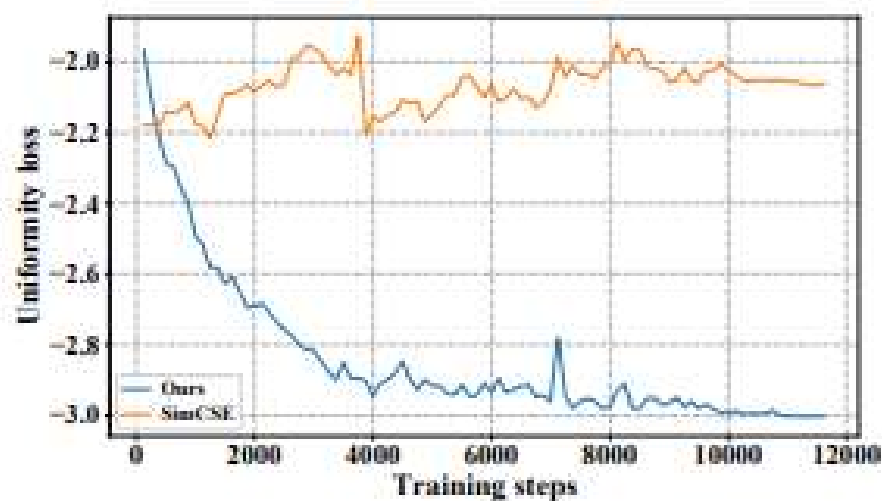


Figure 4: The uniformity loss of DCLR and SimCSE using BERT-base on the validation set of STS-B during training.

$$\ell_{uniform} \triangleq \log \mathbb{E}_{x_i, x_j \stackrel{i.i.d.}{\sim} p_{data}} e^{-2\|f(x_i) - f(x_j)\|^2},$$

04 Experiment

□ Uniformity

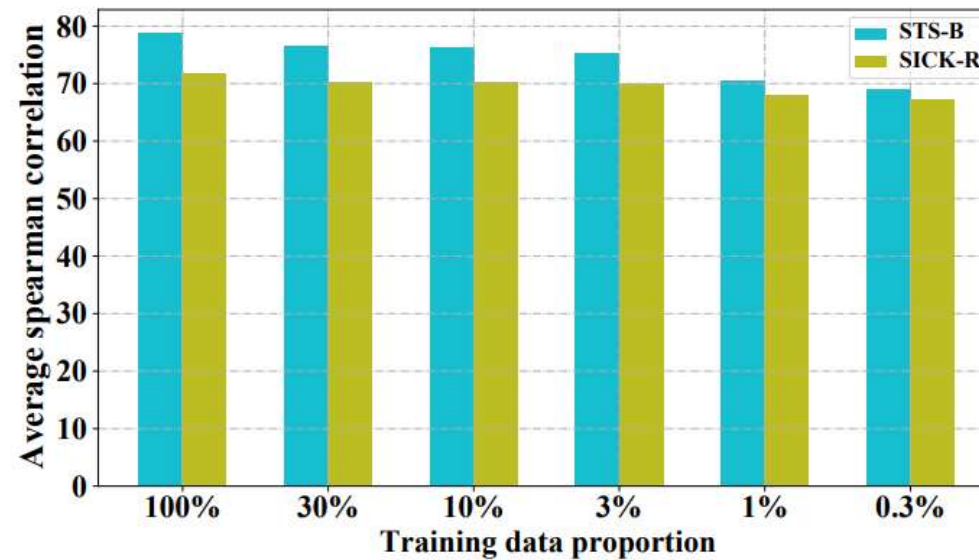
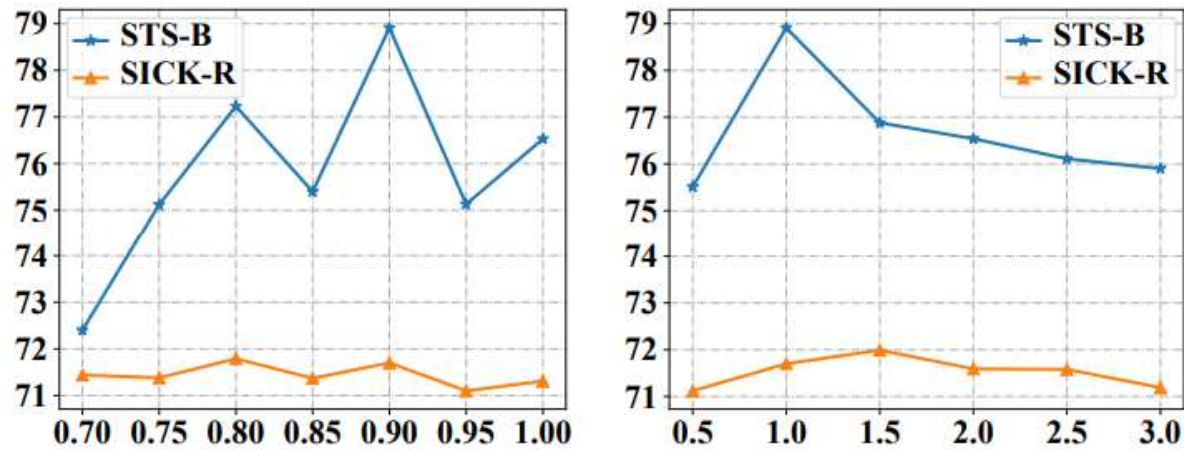


Figure 5: Performance tuning of our DCLR *w.r.t.* different amounts of training data.

04 Experiment

□ Uniformity



(a) Weighting Threshold ϕ (b) Negative Proportion k

Figure 6: Performance tuning *w.r.t.* ϕ and k .

05 Conclusion

- DCLR을 제안하였고 ,

- Core idea

1. Random sampling bias를 줄였다.
2. 그리고 instance weight method를 통해 false negative의 영향을 줄임

- 7 STS tasks에서 효과를 보임



Q&A
